5 **(a)** If the resultant force is directed towards the centre of the circle or perpendicular to the velocity B1

the linear speed remains unchanged but its <u>direction changes with time [B1].</u>

(b) **(i)** $T = Mg = 0.90 \times 9.81 = 9.83$ N A1

Resultant force

$$=\sqrt{T^2-(mg)^2}$$

$$=\sqrt{(8.82)^2-(0.30\times 9.81)^2}$$ M1

A 0

$$=8.3 \text{ N}$$

(ii) Resultant force = $mr\omega^2$

$$9.76 = mr\left(2\pi f\right)^2$$ M1

$$r = \frac{8.324}{0.30\left(2\pi\left(\frac{175}{60}\right)\right)^2} = 0.0826\ m$$ A1

(iii) It would not be possible for the string to be horizontal because the string needs to make an angle with the horizontal such that there is a <u>vertical component of the tension to balance the weight.</u> B1